



Industrial Linux

Security and Robust Filesystem

Keifer Lee

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Keifer Lee

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10+ years

Linux Development Experience

Experience

- Linux development on Android
- Linux development on 5G Small Cell base station
- Linux development on Industrial PC

Education

- National Taiwan University, Master of Department of Communication Engineering

Industrial Linux Must Be...



- **Robust**
- **Secure**
- **Vs. General Linux**

Industrial Linux vs. General Linux

	Industrial Linux	General Linux
Enhanced Security	✓	✗
Stability and Reliability	✓	✗
Regulation Compliance	✓	✗
Industrial Protocol Support	✓	✗
Real-time Capability	✓	✗
New Feature Update	✗	✓
Entertainment Software	✗	✓

Agenda

- **Security**
 - Secure Boot
 - Full Disk Encryption
- **Robust Filesystem**
 - OverlayFS

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Secure Boot



Defined as a security feature that ensures only trusted software is executed during boot process



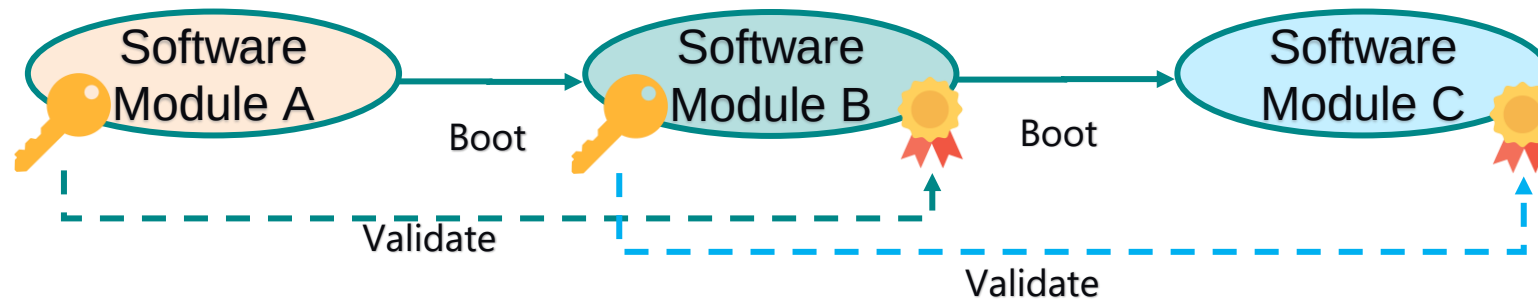
Rootkit => untrusted software

Grant root access, live in boot process, mask existence

- UEFI: Lojax
- Kernel module: Knark

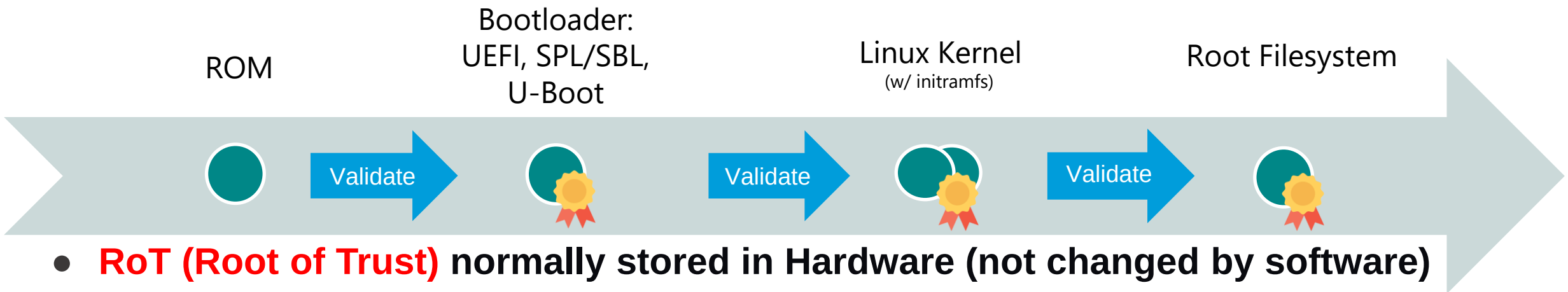
How Secure Boot Works

- Digital signature validation



Secure Boot

- **System boot flow with Secure Boot**



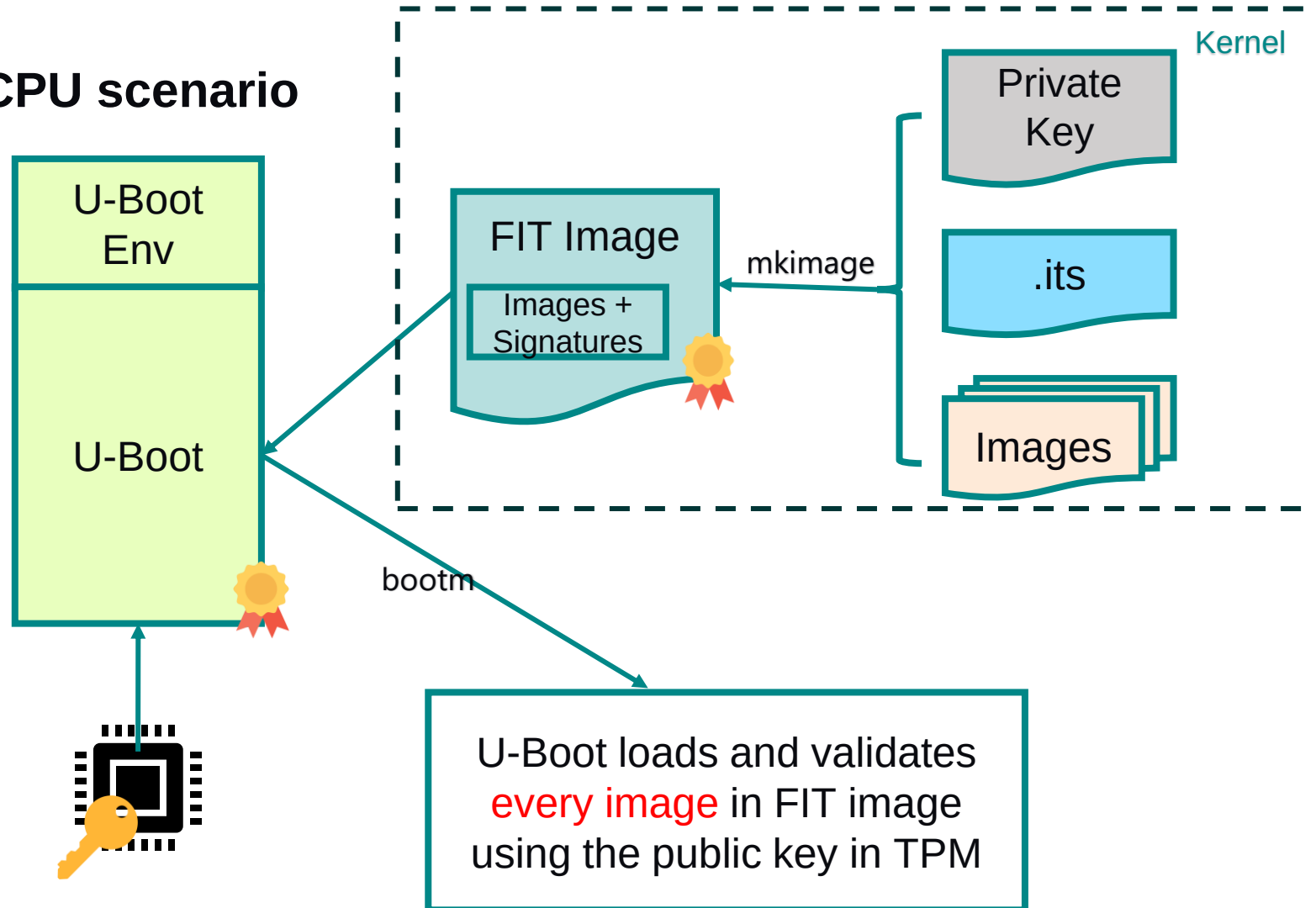
NVRAM (Non-Volatile Random Access Memory)



SOC OTP (One-Time-Programmable)

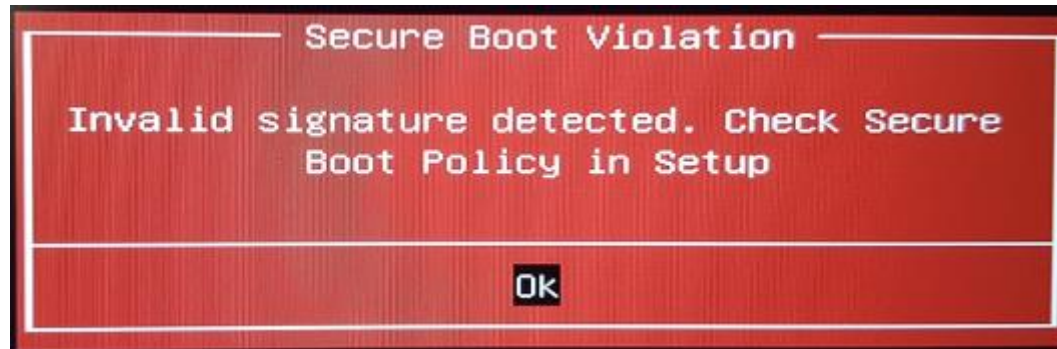
U-Boot Signature Validation

- ARM CPU scenario



U-Boot Signature Validation

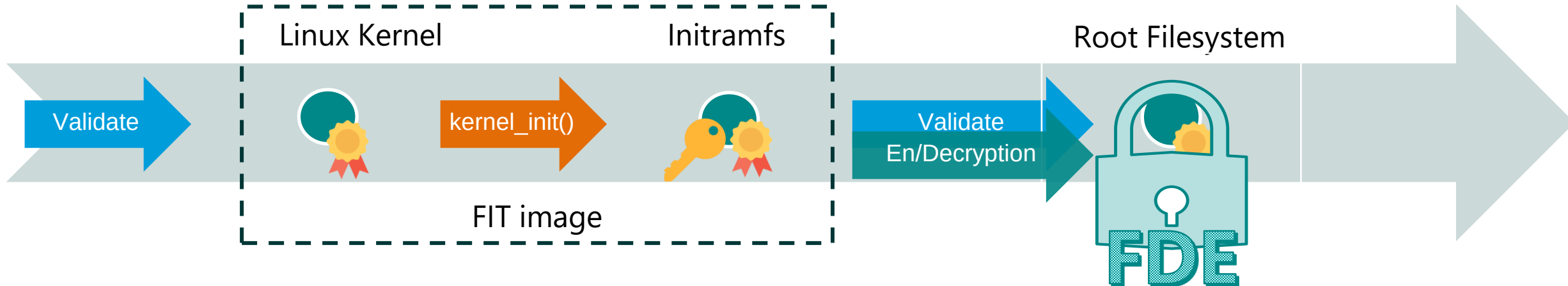
```
## Loading kernel from FIT Image at a6000000 ...
Using configuration
Verifying Hash Integrity ... OK
Trying 'kernel-am64x' kernel subimage
Description: Moxa K3 AM64x Dual Processors Series Un
Created: 2024-08-23 2:55:16 UTC
Type: Kernel Image
Compression: gzip compressed
Data Start: 0xa6000110
Data Size: 8692511 Bytes = 8.3 MiB
Architecture: AArch64
OS: Linux
Load Address: 0x80800000
Entry Point: 0x80800000
Sign algo: sha512,rsa4096:IBG Linux Official TI AM
Sign value: 286141a2cbdbf4c3e894b7c199539a8abbcf8c5
d793e1d3ccda4492181eab6b3bfdd58911a5e547fd315127ce0d292fb5
d1ba39c1e147ef6396f56f50a3946d959c545d5d76619439db4dbb722c
5d785a9bc05c9f1752562c2c2b58494d23692f6c11272d287aa8b9a6c5
cb71afa937e99faa7c916df5581dbacee7a59a4563b010f20ab7bd87db
522b0afd1e05db7f74529ac6696c305fc0d6af44cf1093bfab3a85c3e6
ab79b5248f623acc5559e7ec27916f96b33a992298872fad499cac06cb
ea74348e7aa7ab2deb0c009dcac29d6bc8ce83c0a0fbce40c07fe61f03
af05ec81c0a4ce3decdd0c4f59c805d4d9d950745a9e1b0f0bf7fa5f6a
Timestamp: 2024-08-23 3:06:08 UTC
Authentication passed
```



Root Filesystem Validation

- **Initramfs validate root file-system signature**
 - **Can Linux Kernel validate Root Filesystem directly?**
 - Yes. Use “dm-verity” kernel module! **Android**
 - **Why use initramfs?**
 - More flexible, e.g., companion with Full Disk Encryption

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Agenda

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- Robust Filesystem
 - OverlayFS

LUKS

A standard of Full Disk Encryption



Public Encryption Format

- e.g., LUKS header layout, key slot, encryption parameters...



Compatibility and Interoperability

- LUKS encrypted disk can be apply to any Linux distribution



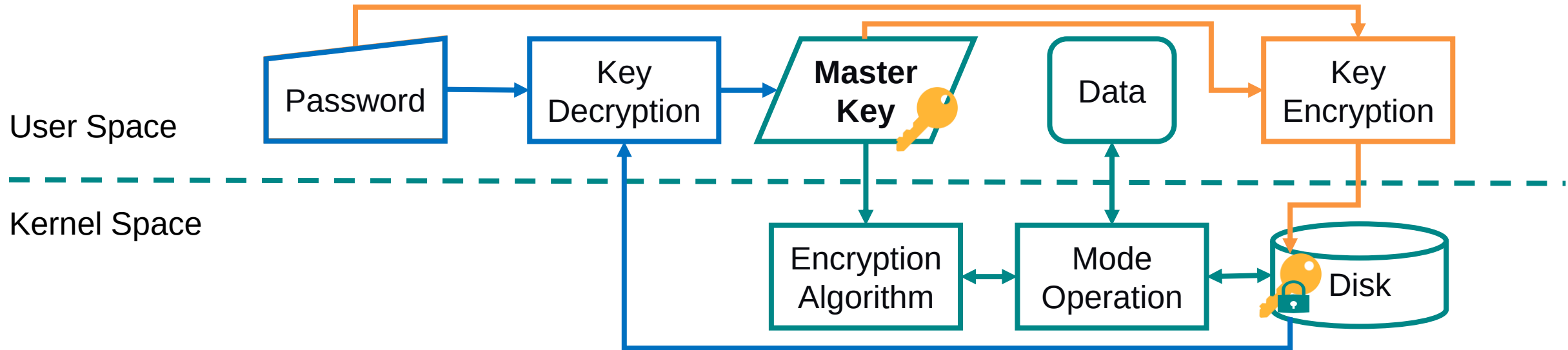
Open Source

- Allow personal and commercial use and publish



- **Master Key**

- **Encrypt full disk/partition**
- **Encrypted and stored at the LUKS header**
- **Decrypted at every boot**
- **Password is needed**

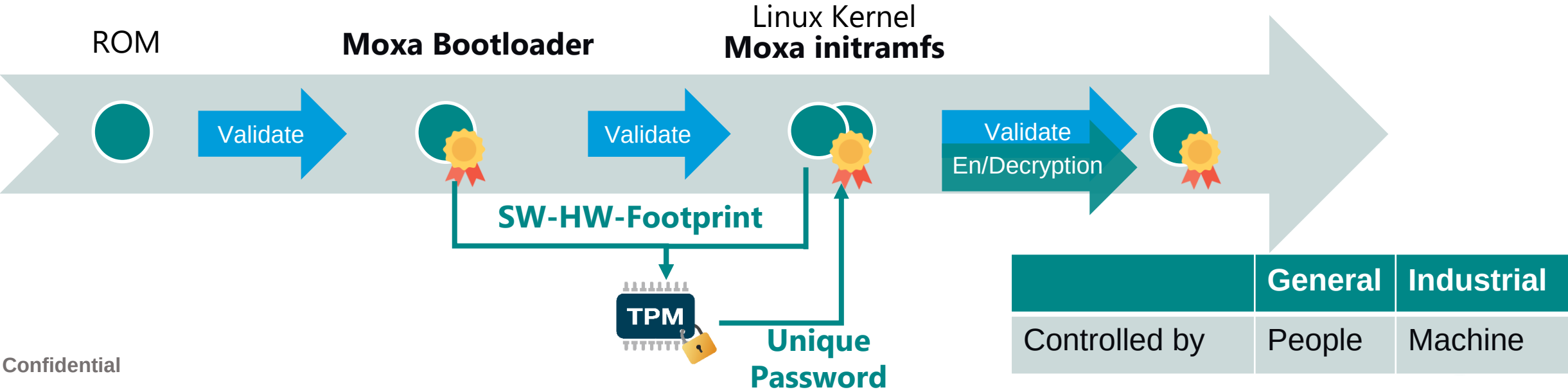


LUKS



Give me auto-encryption/decryption!!

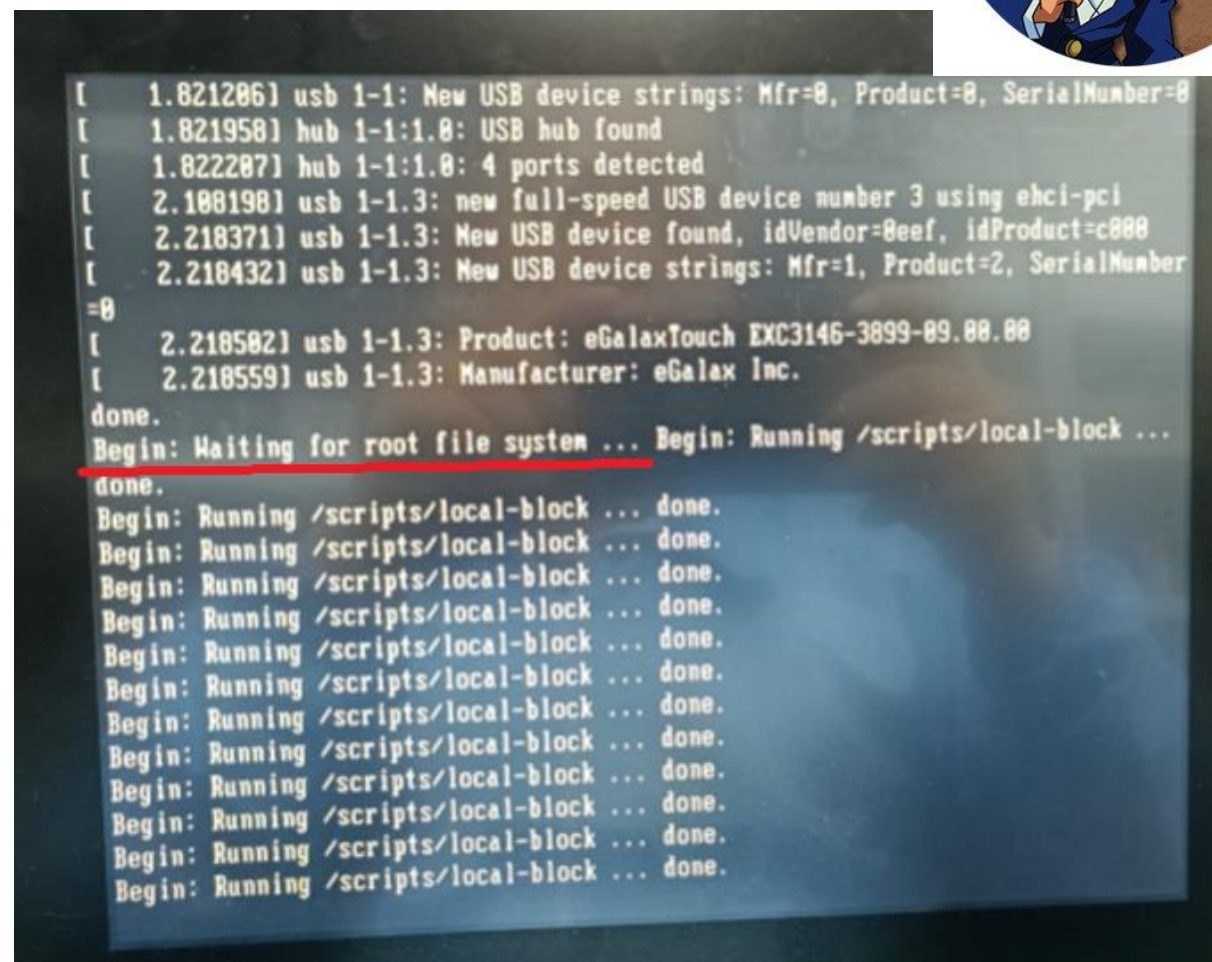
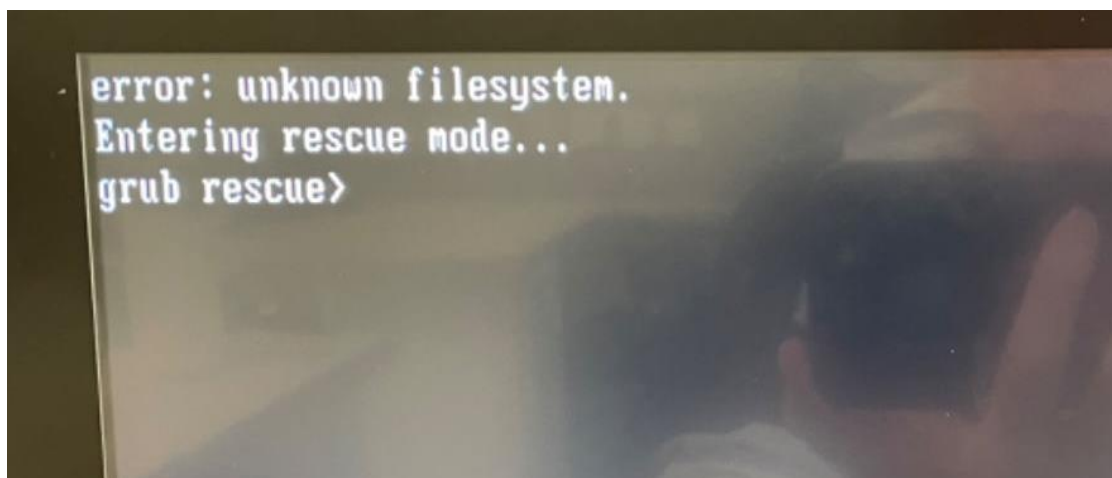
- Moxa Secure Boot + LUKS FDE



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File System Corrupted



File System Corrupted



- Issues happened after power outage
 - Incomplete file operation
- How to fix?



Backup/
Restore

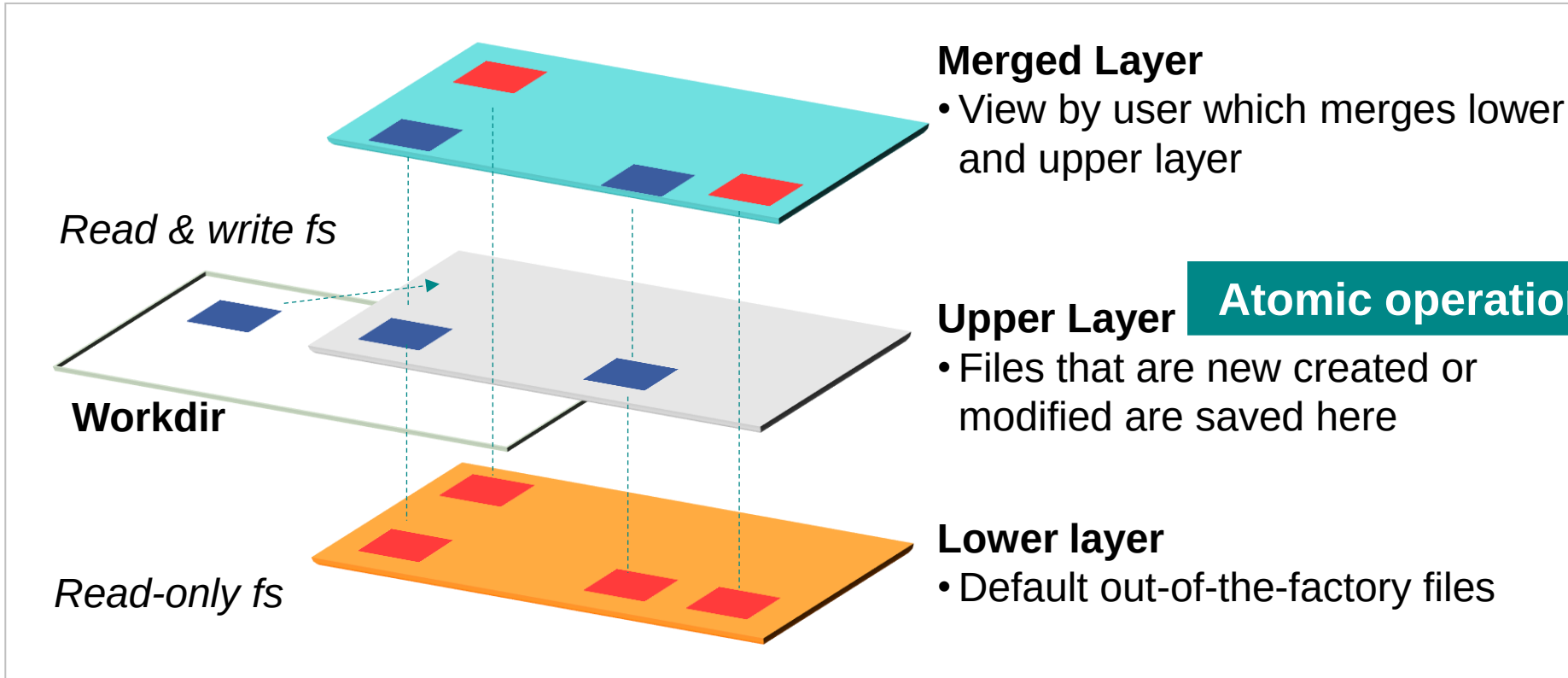
Recovery

Reset
Factory
Default

Format/
Reinstall

**Robust Filesystem -
Overlays**

Overlayfs



Survive **10000+**
power outage tests



Conclusion

- **Industrial Linux needs to be more secure, reliable and stable**
- **Security Boot only allows trusted software boot**
 - Prevent rootkit executed
- **LUKS, a standard of Full Disk Encryption**
 - Prevent important data leaked
- **Robust Filesystem applies overlays**
 - Atomic operation makes filesystem more robust



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- 地點：新莊宏匯廣場(新北大道四段3號13F)
- 請穿著方便行走的鞋子進行實驗室參觀活動
- 因應環保，請自備水壺
- 參訪當日備有午餐

報名截止日11/25(一)
人數有限，不要錯過機會囉！

報名連結



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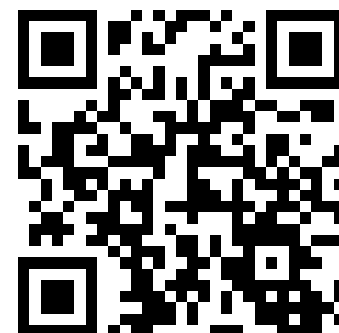
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